Motion on the Unit Circle, Roots of Unity, and Finite Fourier Analysis

N-20220321

§1.1 A Differential Equation for Rotation

Consider the complex-valued initial-value problem

$$Z'(t) = iZ(t), \qquad Z(0) = 1.$$

Its solution is

$$Z(t) = e^{it}.$$

Multiplication by i sends

$$x + iy \longmapsto -y + ix,$$

which is a counterclockwise quarter-turn. Thus the velocity $Z'(t) = iZ(t)$ is always perpendicular to the radius vector.

The equation also proves that the radius is constant. Since

$$|Z|^2 = Z\overline{Z},$$

one has

$$\begin{aligned}\frac{d}{dt}|Z(t)|^2 &= Z'(t)\overline{Z(t)} + Z(t)\overline{Z'(t)} \\ &= iZ(t)\overline{Z(t)} - iZ(t)\overline{Z(t)} = 0.\end{aligned}$$

Because $|Z(0)| = 1$,

$$|Z(t)| = 1$$

for every t. Moreover,

$$|Z'(t)| = |iZ(t)| = 1,$$

so the particle moves around the unit circle with unit speed.

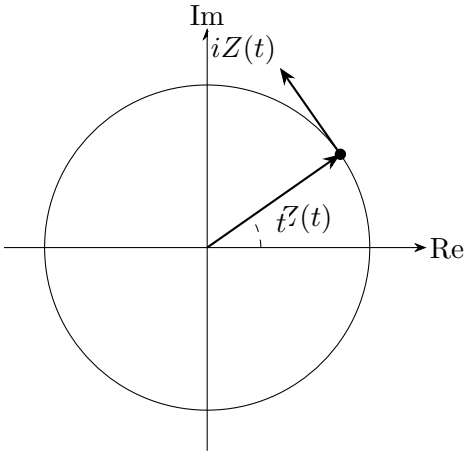


Figure 1.1: The equation $Z' = iZ$ rotates the radius vector by 90°, producing tangent velocity of constant length.

§1.2 The exponential map wraps a line onto a circle

The map

$$\exp_i : \mathbb{R} \longrightarrow S^1, \qquad t \longmapsto e^{it}$$

is a homomorphism from addition to multiplication:

$$e^{i(s+t)} = e^{is}e^{it}.$$

It is surjective because every unit complex number has the form

$$\cos t + i \sin t.$$

Its kernel is

$$\ker \exp_i = 2\pi\mathbb{Z}.$$

Therefore the first isomorphism theorem gives an isomorphism of groups

$$\boxed{\mathbb{R}/2\pi\mathbb{Z} \cong S^1.}$$

Angles differing by a full number of turns are not merely numerically similar; they represent the same coset in the quotient.

Locally, $t \mapsto e^{it}$ is one-to-one, but globally it wraps the real line around the circle infinitely many times. For a continuous loop

$$\gamma : [0,1] \to S^1, \qquad \gamma(0) = \gamma(1),$$

one can choose a continuous lift $\theta : [0,1] \to \mathbb{R}$ with

$$\gamma(t) = e^{i\theta(t)}.$$

The endpoint difference satisfies

$$\theta(1) - \theta(0) = 2\pi k$$

for an integer k, the winding number of the loop. The kernel of the exponential map is what makes this integer possible.

§1.3 Roots and Branch Choices

Let

$$w = \rho e^{i\theta}, \qquad \rho > 0.$$

The equation

$$z^n = w$$

has the n solutions

$$z_k = \rho^{1/n} e^{i(\theta+2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1.$$

The radii are fixed by

$$|z|^n = \rho,$$

while the arguments solve

$$n \arg z \equiv \theta \pmod{2\pi}.$$

The solutions form a regular n-gon on the circle of radius $\rho^{1/n}$.

For example, the fourth roots of -4 are

$$\sqrt{2}\, e^{i(\pi+2\pi k)/4}, \qquad k = 0, 1, 2, 3.$$

They occur at angles

$$\frac{\pi}{4}, \quad \frac{3\pi}{4}, \quad \frac{5\pi}{4}, \quad \frac{7\pi}{4}.$$

The multivalued appearance of complex roots is therefore the geometry of lifting one point through the covering map $z \mapsto z^n$.

§1.4 The roots of unity form a finite cyclic group

Let

$$\zeta_n = e^{2\pi i/n}.$$

The n-th roots of unity are

$$\mu_n = \{1, \zeta_n, \ldots, \zeta_n^{n-1}\}.$$

Multiplication adds exponents modulo n:

$$\zeta_n^a \zeta_n^b = \zeta_n^{a+b}.$$

Hence

$$\boxed{\mathbb{Z}/n\mathbb{Z} \longrightarrow \mu_n, \qquad [k] \longmapsto \zeta_n^k}$$

is a group isomorphism.

The order of ζ_n^k is

$$\frac{n}{\gcd(n,k)}.$$

Indeed, $(\zeta_n^k)^m = 1$ exactly when

$$n \mid km.$$

The smallest positive such m is $n/\gcd(n,k)$.

For $n = 12$, the powers ζ_{12}^k with

$$\gcd(k, 12) = 1$$

are the primitive roots:

$$\zeta_{12}, \quad \zeta_{12}^5, \quad \zeta_{12}^7, \quad \zeta_{12}^{11}.$$

Exact rotational period has become an elementary gcd calculation.

§1.5 Divisibility on Finite Subgroups

Let

$$\omega = e^{2\pi i/3}.$$

The roots of

$$x^2 + x + 1$$

are ω and ω^2. Therefore

$$x^2 + x + 1 \mid x^{2n} + x^n + 1$$

exactly when the latter polynomial vanishes at ω.

If $n \equiv 0 \pmod 3$, then $\omega^n = 1$ and

$$\omega^{2n} + \omega^n + 1 = 3.$$

If $n \equiv 1 \pmod 3$, then

$$\omega^{2n} + \omega^n + 1 = \omega^2 + \omega + 1 = 0.$$

If $n \equiv 2 \pmod 3$, then

$$\omega^{2n} + \omega^n + 1 = \omega + \omega^2 + 1 = 0.$$

Thus

$$\boxed{x^2 + x + 1 \mid x^{2n} + x^n + 1 \iff 3 \nmid n.}$$

This is the first prototype of a character test. Powers of ω remember only the exponent modulo three, so evaluation converts a polynomial identity into arithmetic on a finite cyclic group.

§1.6 Character orthogonality filters one congruence class

For $m \in \mathbb{Z}$, the geometric series gives

$$\sum_{j=0}^{n-1} \zeta_n^{jm} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

Equivalently,

$$\frac{1}{n} \sum_{j=0}^{n-1} \zeta_n^{j(k-r)} = \begin{cases} 1, & k \equiv r \pmod n, \\ 0, & k \not\equiv r \pmod n. \end{cases}$$

This is the roots-of-unity filter.

Let

$$P(x) = \sum_{k \geq 0} a_k x^k.$$

The sum of the coefficients with indices congruent to $r \pmod n$ is

$$\sum_{k \equiv r\,(n)} a_k = \frac{1}{n} \sum_{j=0}^{n-1} \zeta_n^{-jr} P(\zeta_n^j).$$

For example, let

$$P(x) = (1 + x)^8$$

and ask for the sum of coefficients whose exponent is divisible by four. With $\zeta_4 = i$,

$$\begin{aligned} a_0 + a_4 + a_8 &= \frac{1}{4}\big(P(1) + P(i) + P(-1) + P(-i)\big) \\ &= \frac{1}{4}\big(256 + (1 + i)^8 + 0 + (1 - i)^8\big). \end{aligned}$$

Since

$$1 \pm i = \sqrt{2} e^{\pm i\pi/4},$$

one has

$$(1 + i)^8 = (1 - i)^8 = 16.$$

Therefore

$$a_0 + a_4 + a_8 = 72,$$

matching

$$\binom{8}{0} + \binom{8}{4} + \binom{8}{8} = 1 + 70 + 1.$$

The filter replaces a restricted coefficient sum by a short list of evaluations.

§1.7 The DFT as Character Coordinates

For a function

$$f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{C},$$

define

$$\widehat{f}(m) = \sum_{k=0}^{n-1} f(k)\zeta_n^{-mk}.$$

The inverse formula is

$$\boxed{f(k) = \frac{1}{n}\sum_{m=0}^{n-1} \widehat{f}(m)\zeta_n^{mk}.}$$

To verify it, substitute the definition:

$$\begin{aligned}\frac{1}{n}\sum_m \widehat{f}(m)\zeta_n^{mk} &= \frac{1}{n}\sum_m \sum_{j=0}^{n-1} f(j)\zeta_n^{m(k-j)} \\ &= \sum_{j=0}^{n-1} f(j)\left(\frac{1}{n}\sum_m \zeta_n^{m(k-j)}\right) \\ &= f(k).\end{aligned}$$

The roots-of-unity filter is exactly the orthogonality step in this proof.

For $n = 4$, take

$$f = (1, 2, 0, -1).$$

Using $\zeta_4 = i$,

$$\begin{aligned}\widehat{f}(0) &= 1 + 2 + 0 - 1 = 2, \\ \widehat{f}(1) &= 1 + 2(-i) + (-1)i = 1 - 3i, \\ \widehat{f}(2) &= 1 - 2 + 0 + 1 = 0, \\ \widehat{f}(3) &= 1 + 2i - i = 1 + 3i.\end{aligned}$$

Thus

$$\widehat{f} = (2, 1 - 3i, 0, 1 + 3i).$$

Applying the inverse formula recovers all four original entries. The transform is not merely evaluation at roots of unity; orthogonality makes those evaluations a complete coordinate system.

§1.8 Fourier Characters and Circulants

Let S be the cyclic shift on $\mathbb{C}^n$:

$$S(x_0, x_1, \ldots, x_{n-1}) = (x_{n-1}, x_0, \ldots, x_{n-2}).$$

For each m, define the Fourier vector

$$v_m = (1, \zeta_n^{-m}, \zeta_n^{-2m}, \ldots, \zeta_n^{-(n-1)m})^T.$$

Then

$$Sv_m = \zeta_n^m v_m.$$

Hence the Fourier basis diagonalizes the shift.

Every circulant matrix is a polynomial in S:

$$C = c_0 I + c_1 S + \cdots + c_{n-1} S^{n-1}.$$

Therefore

$$Cv_m = \lambda_m v_m,$$

where

$$\lambda_m = c_0 + c_1 \zeta_n^m + \cdots + c_{n-1} \zeta_n^{m(n-1)}.$$

The eigenvalues are the DFT of the first row, up to the chosen sign convention.

For the four-cycle adjacency matrix

$$C = S + S^{-1},$$

the eigenvalues are

$$\lambda_m = \zeta_4^m + \zeta_4^{-m} = 2\cos\frac{2\pi m}{4},$$

so

$$\sigma(C) = \{2, 0, -2, 0\}.$$

A graph operator with coupled coordinates becomes four independent scalar multipliers in the Fourier basis.

Circular convolution is diagonalized for the same reason. If

$$(f * g)(r) = \sum_{k \in \mathbb{Z}/n\mathbb{Z}} f(k) g(r-k),$$

then

$$\widehat{f * g}(m) = \widehat{f}(m)\widehat{g}(m).$$

Convolution in physical coordinates is multiplication in character coordinates.

§1.9 Heat on a Four-Node Cycle

The Fourier basis earns its keep when we let an operator evolve rather than merely diagonalize it. Imagine four sites arranged on a circle, with heat (or a probability distribution) allowed to move between nearest neighbors. If $u_k(t)$ is the quantity at vertex $k \in \mathbb{Z}/4\mathbb{Z}$, the discrete heat equation is

$$\frac{d}{dt} u_k(t) = u_{k-1}(t) - 2u_k(t) + u_{k+1}(t), \qquad u_k(0) = \delta_{k0}.$$

Indices are taken modulo four. In matrix form, $\dot{u} = -Lu$, where

$$L = 2I - S - S^{-1}.$$

The Fourier characters turn the *coupled* system of four differential equations into independent scalar ones. Indeed, for $w_m(k) = \zeta_4^{mk} = i^{mk}$,

$$Lw_m = (2 - \zeta_4^m - \zeta_4^{-m}) w_m = \left(2 - 2\cos\frac{2\pi m}{4}\right) w_m.$$

Thus the four eigenvalues are

$$\lambda_0 = 0, \qquad \lambda_1 = 2, \qquad \lambda_2 = 4, \qquad \lambda_3 = 2.$$

The point source δ_{k0} has Fourier transform $\widehat{u}_0(m) = 1$ for every m. Each coefficient solves $\dot{\widehat{u}}_t(m) = -\lambda_m \widehat{u}_t(m)$, giving

$$\boxed{u_k(t) = \frac{1}{4}\sum_{m=0}^{3} e^{-\lambda_m t}\, i^{mk}.}$$

Returning to real coordinates and writing $a = e^{-2t}$, we get the exact four-site solution

$$u_0(t) = \frac{1 + 2a + a^2}{4}, \quad u_1(t) = u_3(t) = \frac{1 - a^2}{4},$$
$$u_2(t) = \frac{1 - 2a + a^2}{4}.$$

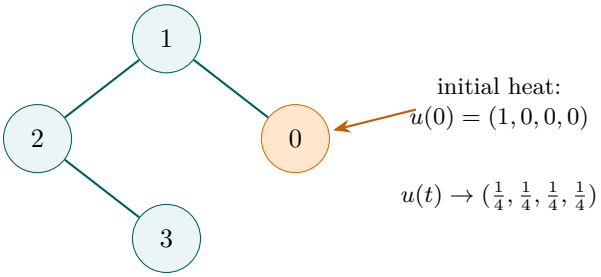


Figure 1.2: The four-cycle has one constant Fourier mode and three decaying modes. Mass spreads from the highlighted vertex until the distribution becomes uniform.

These expressions are automatically nonnegative for $t \geq 0$ because $0 \leq a \leq 1$, and they sum to one:

$$\sum_{k=0}^{3} u_k(t) = 1.$$

At $t = 0$, $a = 1$ and all heat starts at vertex zero. As $t \to \infty$, the nonconstant Fourier modes decay and each site approaches $1/4$. The surviving zero-eigenvalue mode represents conservation of total mass, while the first nonzero eigenvalue 2 controls the slowest exponential relaxation. This is the spectral gap of the graph Laplacian.

The computation has unified three pictures with no extra black box: roots of unity are characters of a cyclic group, characters diagonalize circulant matrices, and those same eigenvectors solve heat flow and continuous-time random walks on a periodic graph.[1]

§1.10 Cyclotomic factors separate roots by exact period

The factorization

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x)$$

groups roots of unity according to exact order. For $n = 12$,

$$\begin{aligned} x^{12} - 1 &= \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x) \\ &= (x-1)(x+1)(x^2+x+1)(x^2+1) \\ &\quad \cdot (x^2 - x + 1)(x^4 - x^2 + 1). \end{aligned}$$

[1] For the diagonalization of circulant matrices by the Fourier matrix, see Gilbert Strang, MIT 18.065, Lecture 31.

The last factor contains exactly the primitive twelfth roots. This is enough for the present chapter: finite Fourier analysis uses all characters of the cyclic group, while cyclotomic factorization separates those characters by exact order. The arithmetic of the factors themselves belongs to the dedicated cyclotomic chapter.

§1.11 Fifth Roots and the Pentagon

Let

$$\zeta = e^{2\pi i/5}$$

and set

$$u = \zeta + \zeta^{-1} = 2\cos\frac{2\pi}{5}.$$

Because

$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0,$$

dividing by ζ^2 gives

$$\zeta^2 + \zeta + 1 + \zeta^{-1} + \zeta^{-2} = 0.$$

Now

$$\zeta^2 + \zeta^{-2} = u^2 - 2,$$

so

$$u^2 + u - 1 = 0.$$

The positive root is

$$u = \frac{\sqrt{5} - 1}{2}.$$

Therefore

$$\cos\frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}.$$

The central angle of a regular pentagon is obtained using only a square root, which is why the pentagon is constructible by straightedge and compass.

This small calculation is the appropriate arithmetic endpoint here. The finite cyclic group supplies the roots, Fourier characters use all their powers as coordinates, and the real combination $\zeta + \zeta^{-1}$ collapses a complex symmetry orbit to the quadratic number controlling the pentagon.